Venkata Lakshmi Parimala Pasupuleti

Senior Data + AI Engineer · Python · PHP · ETL/ELT · Schema Design · RAG · LLM Evaluation · AWS · AI-Assisted Development · Real Estate Finance · Production AI

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PROFESSIONAL SUMMARY

Data + AI engineer with 8 years shipping production ETL/ELT pipelines, financial data schemas, and LLM-powered systems. At Entrata: **schema design and normalization** for financial operations — HAP allocation, rent schedules, gross rent, HUD payments, tax credits, income limits, and distribution analytics across 20+ production modules (1M+ records, 100K+ MAU, AWS S3/RDS/Glue); processed 1.6M HUD/Tax Credit income limit records; migrated client document storage to AWS S3 (90% legacy space constraints resolved); 56 state-specific certification documents. Built **Leasel**: agentic RAG pipeline (93.6% precision) with **deterministic checks, cross-model comparison, and human-in-the-loop review**. Built **AgentProbe**: 5-metric AI output evaluation engine with cross-model LLM comparison and CI/CD regression gates. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — ETL Pipelines · ML Modeling · Analytics Engineering · Observability

- Built multi-source ETL pipelines (Python, R, 190+ countries, WHO + UK Road Safety datasets) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; designed analytics dashboards (Tableau, ggplot2, Quarto) driving **40% improvement** in business decision comprehension for faculty leadership.

Entrata — Real Estate PropTech SaaS (Opendoor Competitor)

Sep 2018 – Aug 2024

Senior Software Engineer — ETL/ELT Pipelines · Schema Design · Financial Data · Production AI · AWS

- Designed and owned **ETL/ELT pipelines, schema normalization, and investment-level financial data models** for property owners and fund managers — HAP allocation, rent schedules, gross rent updates, HUD payment processing, tax credit calculations, ledger management, income eligibility, and housing subsidy compliance across 20+ production modules (**1M+ records, 100K+ MAU**, 99.5% uptime, 120+ Agile sprints).
- Processed **1.6M HUD/Tax Credit income limit records** via production AI pipeline (Python, PyTorch, OpenAI API) replacing manual Excel workflows; led **database architecture decisions** for complex production schemas; **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, partitioning, caching, triggers, log tables, and audit tables; led microservices Docker redesign (50% scalability improvement, 30% deployment reduction).
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium encryption (100% of resident data encrypted, 50%+ of codebase updated); migrated thousands of client documents to AWS S3 resolving 90% of legacy space constraints; created **56 state-specific certification documents** with AWS S3 integration; eliminated 80% manual workload via Python/Bash automation; **300+ code reviews, 5+ engineers mentored**.

Entrata India

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · Schema Design · AWS S3

- Built full-stack property management modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts for 1.6M HUD/Tax Credit income limit records; developed data normalization pipelines, compliance workflows, and AWS S3 document storage integrations; participated in sprint planning, code reviews, and performance debugging.

PROJECTS

Leasel: Agentic RAG — Deterministic Checks, Cross-Model Comparison, Human-in-the-Loop Eval

PROPTECH · AI EVALUATION

Python · LangGraph · ChromaDB · pgvector (familiar) · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 residential lease PDFs — agentic RAG pipeline with vector store retrieval and LLM reasoning over real estate legal text; **deterministic checks + cross-model LLM comparison + human-in-the-loop review** eval harness (pass@1 81.7%, pass@3 88.3%); CI/CD regression gate blocks regressions automatically.

AgentProbe: AI Output Evaluation with Cross-Model Comparison & Observability Dashboard

AI-ASSISTED DEV · LLM EVALUATION

Python · LangGraph · Claude API · OpenAI · Groq · Ollama · FastAPI · ChromaDB · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- **5-metric AI output evaluation engine** with **cross-model comparison** (Claude, OpenAI, Groq, Ollama), deterministic checks, CI/CD regression gates on every commit; React **observability dashboard** tracks per-run metrics, retrieval drift, and latency; 3-agent LangGraph runtime with tool calling and inter-agent state handoffs.

CrashClarity: Analytics Engineering on 104K Road Safety Records

ANALYTICS ENGINEERING

Python · R · scikit-learn · Random Forest · Feature Engineering · ETL · Tableau · GitHub Pages — Portfolio

- **85%+ accuracy** on 104K+ UK accident records — ETL pipeline, schema normalization, feature engineering (weather, road, time), Random Forest + Logistic Regression, interactive observability dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Data engineering, ML systems, statistical modeling, AI pipelines, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Data Engineering & Backend: Python (expert), PHP, SQL (expert — 200+ production optimizations, schema design, normalization, execution plan tuning, indexing, partitioning, triggers, audit tables), PL/SQL, JavaScript, TypeScript, R; ETL/ELT Pipelines, Schema Design, Data Normalization, Schema Mapping, Data Validation, Observability Tooling; AWS (S3, EC2, RDS, Glue), PostgreSQL, MySQL, Docker, Libsodium, CI/CD (GitHub Actions), Microservices, Git

AI & LLM Evaluation: LangGraph, LangChain, RAG Pipelines, Vector Databases, ChromaDB, pgvector (familiar), OpenSearch (familiar), Agentic Workflows, AI-Assisted Development, AI Output Evaluation, Deterministic Checks, Cross-Model Comparison, Human-in-the-Loop Review, LLM-as-Judge, Eval Frameworks, Prompt Engineering, Claude API, OpenAI API, Groq, Ollama, FastAPI, NLP, BERT

ML & Analytics: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Anomaly Detection, Statistical Modeling, Tableau, Power BI, ggplot2, Agile/Scrum, Planning Poker